Carbon:

+ Diamond (sp^3), Graphite (sp^2) → unhybridized p-orbitals are perpendicular to the hexagonal ring structure ↓ participate in π bonding (w/ density delocalized over the layers)

+ @ RTP, graphite is thermodynamically more stable than diamond.

This weak interaction allows the layers to slip ↓ Good lubricant & electrical conductor

+ Hydrogen actually bonds to surface C in diamond to reduce its coefficient of friction

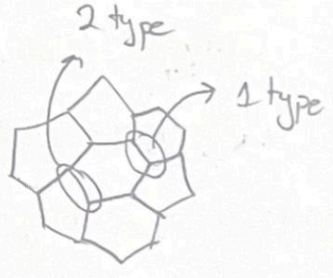
+ Graphene → one layer of graphite, resistant to fracture, deformation, high thermal conductivity, (zero band gap)

+ Nanoribbons/ Nanocarbons: formed via lithographic techniques (cutting up & folding graphene)
Types: zigzag, armchair, chiral

↳ Conductivity of carbon nanotubes varies w/ diameter of the tubes & their chirality (can be semiconductors or super conductive)
↳ can be unzipped using $KMnO_4$

+ Graphyne: carbon-triple bond planar structure

+ Fullerenes: C_{20}, C_{60}, C_{70}



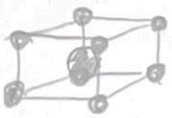
↓
→ fused 5 & 6 membered C rings
→ each pentagon is fused to 5 hexagons
→ each hexagon is alternatively surrounded by hexagons and pentagons
→ Two types of C-C bond occur

Bohr $E = h\nu = \frac{hc}{\lambda}$; $E = KE + \phi$; $n\lambda = 2d\sin\theta$ (diffraction) ↑ ϕ = Work function

MSE stuff

Cubic

↳ 14 Bravais lattices: →

	Simple	BCC	FCC
			
No. of atoms	1	2	4
(Coord. no.)	6	8	12

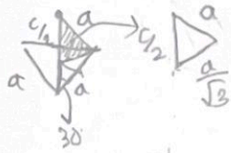
HCP



6 atoms

12 coord. no's.

$a = 2R$



$\frac{c^2}{2} = a^2 - \frac{a^2}{3}$

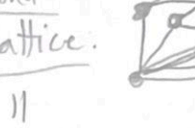
$= \frac{2a^2}{3}$

$c = 2a\sqrt{\frac{2}{3}}$



$\cos 30 = \frac{a/2}{x}$

$x = \frac{a/2}{\cos 30} = \frac{a}{\sqrt{3}}$

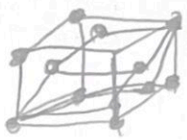


$a = \frac{4R}{\sqrt{3}}$



$a = 2R\sqrt{2}$

Diamond lattice.

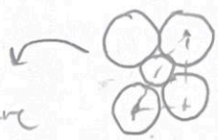




||

Zinc Blende

(But diff atoms)



$CsCl$ → BCC structure



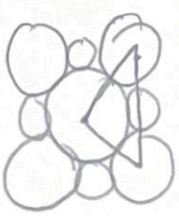
Wurtzite → ZnS ($\frac{1}{2}$ T_d sites in FCC)

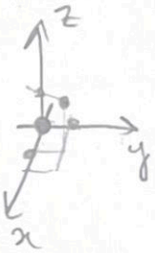
Fluorite → CaF_2 (occupying T_d sites in FCC)

Rutile → TiO_2

Perovskite → $CaTiO_3$

NaCl → FCC (All O_h sites are occupied)







[Lecture] 104A: Group Theory!

Schoenflies Symbols

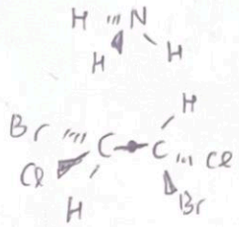
C_{2v}, D_{3h}, T_d ....

Case

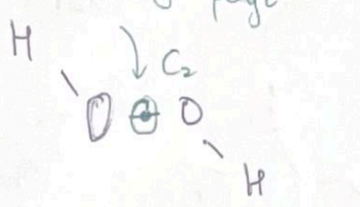
(1) Only E → C_1

(2) Only E, σ → C_s → $S_1 \equiv \sigma$

(3) Only E, i → C_i



Coming out of page



$\{E, C_2, \sigma_h, i\}$

C_{2h}

mirror plane which the molecule lies in

C_{2h} usually always has an i

(4) Rotational Groups (C_n, C_{nh}, C_{nv})

C_{nh} → C_n and add σ_h

C_{nv} → C_n & add σ_v (vertical mirrors)

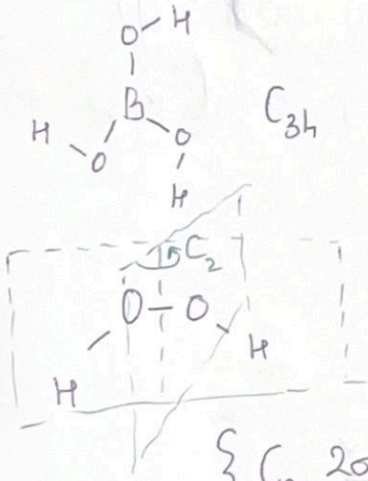
(5) Dihedral Groups (D_n, D_{nh}, D_{nd})

Multiple C_n axes ($\perp C_2$'s)

(i) D_n: $n C_2$ axes $\perp$ to C_n

(ii) D_{nh}: " ; add σ_h

(iii) D_{nd}: " ; add σ_v which are called σ_d

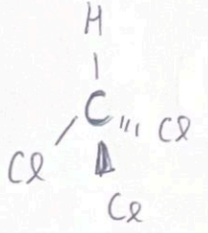


C_{3h}

$\{C_2, 2\sigma_v, E\}$

$\Downarrow$

C_{2v}

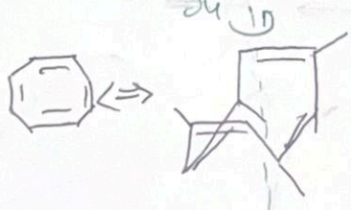


C_{3v}

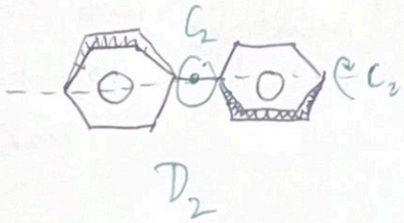
(6) S_n groups

(i) S_n, n even $S_2 \equiv i \Rightarrow C_i$

S_4, S_6, S_8 .... n elements S_n



(ii) n odd ⇒ have σ_n ⇒ D_{nh}



D_2

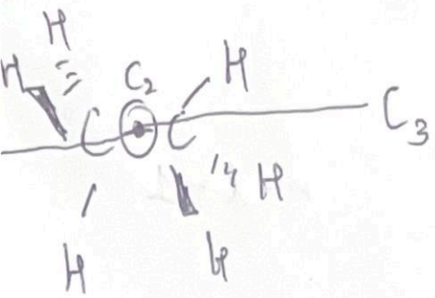
(7) Linear Groups ($C_{\infty v}$, $D_{\infty h}$)

$C_{\infty v}$: ∞ $\perp C_2$'s

$D_{\infty h}$: $\perp C_2$'s

$-C\equiv O-$ C_∞

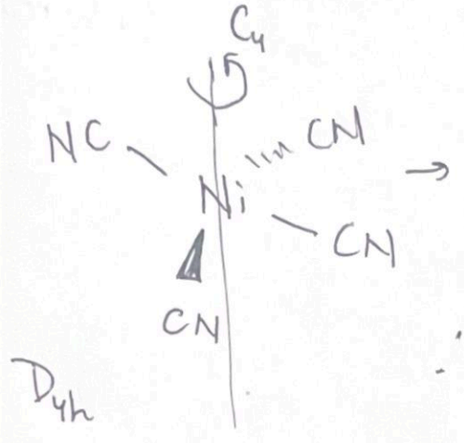
HCl, HCN



† If C_2 & C_3 are perpendicular then there have to be 2 C_3's and 3 C_2's

† Principle axes of rotation ⇒ Highest order.

$2C_2'$ → prime



D_{4h}

C_4 twice → C_4^2, thrice → C_4^3

C_4, C_4^2, C_4^3

C_4^2 → Same as C_2

∴ We have $2C_4$ → C_4, C_4^3

∴ C_4 will always have $2C_4$ & C_2 ⇒ But in exam write down every single one list & the total no.

$\{E, 2C_4, C_2, 2C_2', 2C_2'', i, 2S_4, \sigma_h, 2\sigma_v, 2\sigma_d\}$

$2C_4$ → On same axis; $2C_2'$, $2C_2''$ → On different axis

$2C_2''$ This is called a class because you can always get C_2'' from the other C_2'' by any symmetry operation →

However there is no way of going from C_2' to C_2'' via any of the symmetry in the whole D_{4h} group.

A good way to see if element symmetries are same is by drawing a vector on each molecule and seeing how the vector changes → Therefore $S_4' \neq C_4'$ (orbitals & vectors are changing)

Orbitals & Drawing Shit:

+ All about Nods: Angular (planar) & Radial.

↳ $N = n - 1$

Total nodes

↳ Angular nodes = l

$s = 0$
$p = 1$
$d = 2$
$f = 3$
⋮

↳ radial nodes = $n - l - 1$

Know the Shapes of Orbitals!

As you cross a node → there is phase/sign change. (1 node)

Electronic Config

↓

for Lanthanides

